GEOTECHNICAL ENGINEERING

UNIT - II : PERMEABILITY & SEEPAGE THROUGH SOILS

Course Code: 23A01503T

Syllabus: PERMEABILITY: Soil water - Capillary rise - Flow of water through soils - Darcy's law - Permeability, factors affecting - Laboratory determination of coefficient of permeability - Permeability of layered systems. SEEPAGE THROUGH SOILS: Total, neutral and effective stresses - Quick sand condition - Seepage through soils - Flow nets: characteristics and uses.

1. Soil Water

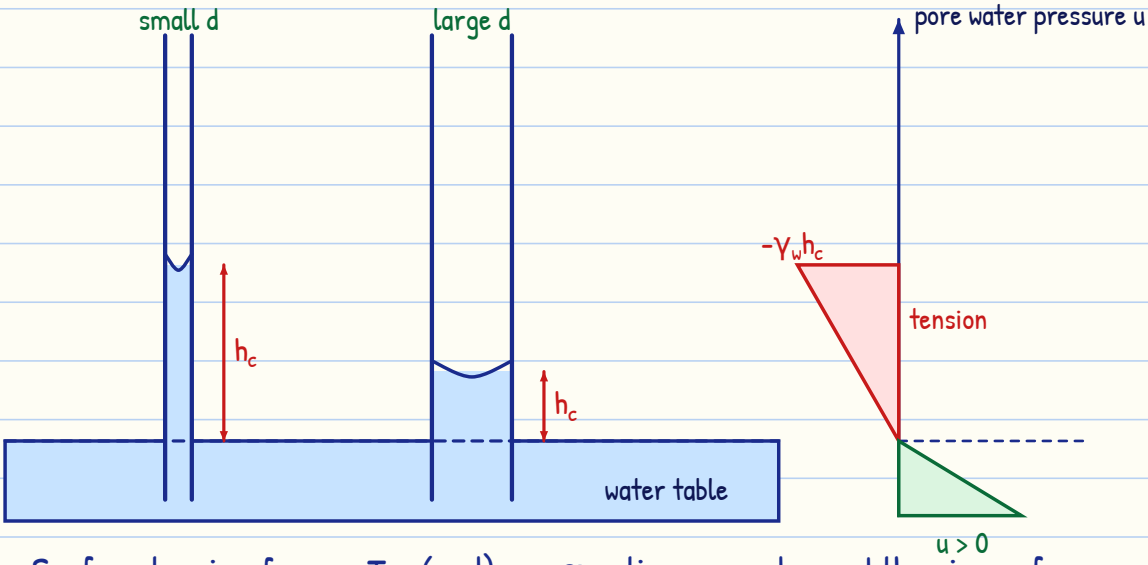
Water in the soil can be classified according to how it is held in the voids and on the grains.

Type	Description
Free (gravitational) water	Moves under gravity, drainage; found below the water table
Capillary water	Held in the fine voids above water table by surface tension
Adsorbed (hygroscopic) water	Held on clay surfaces by electro-chemical forces; not removable by gravity
Structural (crystal) water	Part of the crystal lattice of minerals, removed only at very high temperatures

- Ground water table (WT): the surface at which the pore water pressure is equal to atmospheric ($u = 0$). Below it, soil is saturated.
- Phreatic surface: the top flow line of seepage in an earth dam; same idea as the WT.
- Perched water table: local water table above an impermeable lens, above the main water table.
- Artesian condition: water in a confined aquifer under pressure greater than atmospheric (head above the ground level).

2. Capillary Rise

Water rises above the water table in a fine tube (or in the fine pores of soil) due to surface tension T of water and the adhesion to the tube walls.



- Surface tension forces: $T \times (\pi\ d) \cos\alpha$ acting upward round the circumference. Weight of the lifted water column: $(\pi\ d^2/4)\ h_c\ \gamma_w$.

$$h_c = 4\ T \cos\alpha\ /\ (\gamma_w\ d)$$

- T = surface tension of water = 0.073 N/m (at 20 degree C); α = contact angle (0 degree for clean glass and water); d = diameter of tube. For water: h_c (cm) = 0.3 / d (cm).
- In soils the equivalent pore diameter is about $D_{10}/5$, so: $h_c = C / (e\ D_{10})$, where C = 0.1 to 0.5 cm^2 (Hazen / Terzaghi-Peck).

Points to remember

- Smaller the pore (finer the soil), higher the capillary rise, but the rate of rise is slower.
- Above the water table, the pore water pressure is negative (tension): $u = -\gamma_w\ h$ at height h above WT. At the top of the capillary fringe $u = -\gamma_w\ h_c$.
- Capillary fringe may be saturated or partially saturated; fully saturated zone (about 2/3 h_c) then partially saturated zone above.

Soil	Coarse sand	Fine sand	Silt	Clay
Approx. capillary rise	0.03 - 0.12 m	0.3 - 3.5 m	1.5 - 12 m	2 - 50 m

Effects / importance of capillarity

- Negative pore pressure raises the effective stress, giving apparent cohesion (damp sand holds a sand castle; dry or fully submerged sand has none).
- Bulking of sand: moist sand (water content 4 - 6 %) occupies up to 20 - 30 % more volume, since capillary films keep the grains apart.
- Frost heave: capillary water feeds ice lenses and heaves pavements in cold regions.
- Capillary water makes foundations, basements and road subgrades damp (damp-proof course is needed); shrinkage of clays is due to capillary tension.

Example

For a soil, D_{10} = 0.02 mm = 0.002 cm, e = 0.60, C = 0.3 cm^2.

h_c = 0.3 / (0.60 x 0.002) = 250 cm = 2.5 m.

3. Flow of Water Through Soils

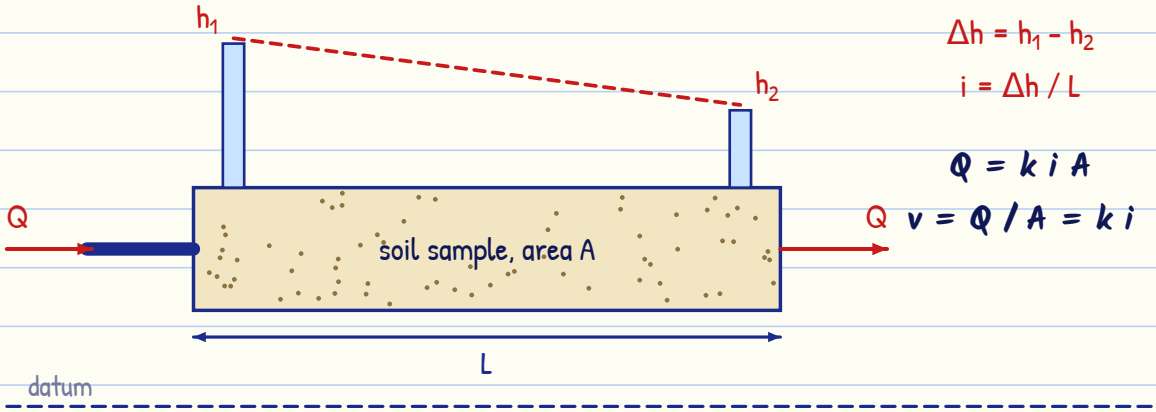
Water flows through the interconnected voids. The driving force is the difference in total head between two points.

- By Bernoulli's equation the total head $h = (u / \gamma_w) + z + (v^2 / 2g)$. In soils the velocity v is very small, so the velocity head is neglected:

$$h = u / \gamma_w + z \quad \text{(pressure head + elevation head)}$$

- Water always flows from higher total head to lower total head, not necessarily from high pressure to low pressure.
- Hydraulic gradient $i = \Delta h / L$ = head loss per unit length of flow path (dimensionless).
- Flow is laminar in soils (Reynolds number $Re = v D \rho / \eta < 1$, in most soils; it can be turbulent in coarse gravel / rock fill).

Darcy's Law (1856)



Henry Darcy, studying water filters at Dijon (France), found experimentally that for laminar flow through saturated soil the discharge velocity is directly proportional to hydraulic gradient.

$$v = k i \qquad q = Q = k i A$$

- v = discharge (superficial) velocity = Q/A; k = coefficient of permeability (hydraulic conductivity) with the dimensions of velocity (cm/s, m/s, m/day); A = total cross-section area of soil (solids + voids).

- Coefficient of permeability k: the discharge velocity under unit hydraulic gradient (i = 1).
- The real velocity in the pores is the seepage velocity v_s. Flow area is only the void area $A_v = n A$, therefore

$$v_s = v / n = v (1 + e) / e \qquad (v_s > v)$$

Validity of Darcy's law

- Valid for laminar flow (Re < 1; up to 10 for practical use), saturated soil, steady flow, incompressible soil and water.
- Not valid in very coarse gravels / rock fill (turbulent), and in very fine clays at very low gradients (below a threshold gradient).

Typical values of k

Soil	Clean gravel	Coarse sand	Fine sand	Silty sand	Silt	Clay
k (cm/s)	> 1	$1 - 10^{-2}$	$10^{-2} - 10^{-3}$	$10^{-3} - 10^{-5}$	$10^{-5} - 10^{-7}$	$< 10^{-7}$
Drainage	very good	good	good	poor	poor	practically impervious

k of clean sand is about a million times that of clay. Remember: k is a property of both the soil and the fluid. $k = K \gamma_w / \eta$, where K = intrinsic (absolute) permeability (cm^2).

4. Factors Affecting Permeability

- Grain size: k increases with the square of the particle size. Hazen: $k = C D_{10}^2$ (k in cm/s, D_{10} in mm, C = 100 for loose clean sand, range 100 - 150).
- Void ratio: larger e gives larger k. Different relations: k proportional to e^2, or $e^3/(1+e)$ (Kozeny-Carman). For sands $k_1 / k_2 = [e_1^3/(1+e_1)] / [e_2^3/(1+e_2)]$.
- Shape and texture of the particles: angular grains give a more tortuous flow path and lower k than rounded grains; rough surface lowers k.
- Structure and stratification: flocculated clays have higher k than dispersed; k parallel to the bedding is more than k perpendicular ($k_h > k_v$).
- Degree of saturation: partly saturated soil (air bubbles block pores) has lower k, so use fully saturated, de-aired water in tests.
- Properties of the permeant: k is proportional to γ_w / η. Viscosity falls as temperature rises so k increases with temperature. Results are reported at 27 degree C (IS) or 20 degree C.

$$k_{27} = k_T \times (\eta_T / \eta_{27})$$

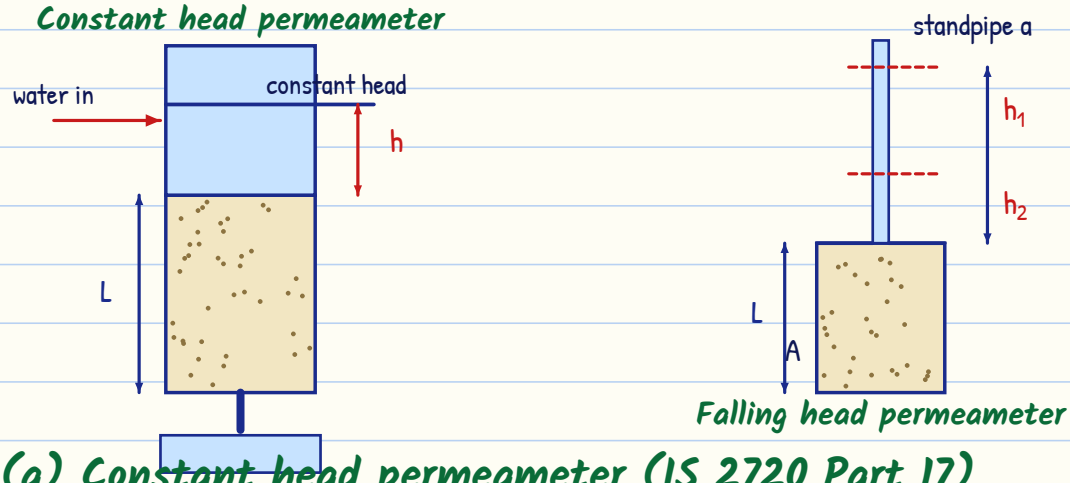
- Adsorbed water in clays blocks the pore throat, lowering k. Impurities / organic matter / entrapped air also reduce k.
- Casagrande: for fine sands, $k = 1.4\, e^2\, k_{0.85}$ ($k_{0.85}$ is k at e = 0.85).

Example (effect of void ratio)

For a sand, $k = 4 \times 10^{-2}$ cm/s at e = 0.60. Find k at e = 0.80.

$k_2 = k_1 \times [e_2^3/(1+e_2)] / [e_1^3/(1+e_1)] = 4 \times 10^{-2} \times (0.512/1.8) / (0.216/1.6) = 4 \times 10^{-2} \times 0.2844 / 0.135 = 8.4 \times 10^{-2}$ cm/s.

5. Laboratory Determination of Coefficient of Permeabi



(a) Constant head permeameter (IS 2720 Part 17)

Suitable for coarse-grained soils (clean sand and gravel) having $k > 10^{-3}$ cm/s. Head h is maintained constant, so the discharge Q is steady.

- Soil sample of length L and cross-sectional area A is placed in a cylindrical mould (100 mm dia, 127.3 mm long); water is passed under constant head h, and the quantity Q collected in time t is measured.

$$k = Q\,L / (A\,h\,t)$$

- With a constant head, the water collected in time t equals $Q = k\,(h/L)\,A\,t$.

Example 1

A sample 15 cm long and 50 cm^2 in area under a constant head of 40 cm gave 450 cc of water in 2 minutes. Find k.

$k = (450 \times 15) / (50 \times 40 \times 120) = 6750 / 240000 = 0.028$ cm/s.

(b) Falling head permeameter (IS 2720 Part 17)

Suitable for fine-grained soils (silt and clay), $k < 10^{-3}$ cm/s. Water flows from a standpipe of area a through the sample; head falls from h_1 to h_2 in time t.

- At any instant, rate of fall: $-a\,dh/dt = k\,(h/L)\,A$. Integrating from h_1 to h_2:

$$k = (a\,L / A\,t)\,\ln(h_1 / h_2) = 2.303\,(a\,L / A\,t)\,\log_{10}(h_1 / h_2)$$

Example 2

Falling head test: $a = 0.5$ cm^2, $A = 30$ cm^2, $L = 12$ cm, $h_1 = 100$ cm, $h_2 = 60$ cm, $t = 10$ minutes.

$k = [0.5 \times 12 / (30 \times 600)] \times \ln(100/60) = 3.33 \times 10^{-4} \times 0.5108 = 1.7 \times 10^{-4}$ cm/s.

Comparison

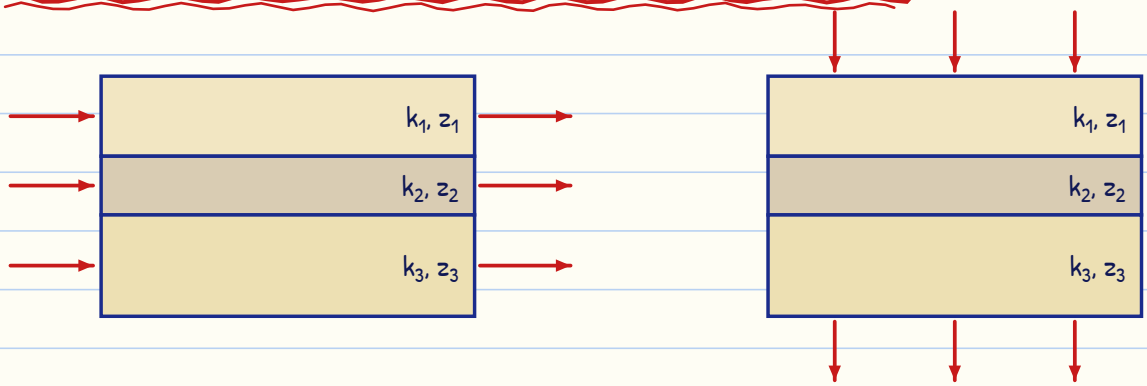
Constant head	Falling head
coarse soils, $k > 10^{-3}$	fine soils, $k < 10^{-3}$
head constant, Q measured	head falls, time measured
quick, reliable for sands	accurate for silts / clays

(c) Other methods

- Indirect methods: from grain size (Hazen's formula), from the consolidation test ($k = c_v\,m_v\,\gamma_w$) and from capillarity test (horizontal).
- Field methods: pumping-out tests (for unconfined aquifer $k = Q\ln(r_2/r_1) / [\pi\,(h_2^2 - h_1^2)]$; for confined aquifer $k = Q\ln(r_2/r_1) / [2\pi\,D\,(h_2 - h_1)]$), borehole tests, tracer / slug tests.
- Field tests give more representative value as they include layering and fissures.

In a laboratory test the sample should be fully saturated and flow upward so that air is carried out; keep the hydraulic gradient low to avoid piping in loose sand.

6. Permeability of Layered Systems



Flow parallel to layers *Flow perpendicular to layers*

Natural deposits are stratified, so the equivalent permeability depends on the direction of flow. Let layers have permeabilities $k_1, k_2, \ldots k_n$ and thicknesses $z_1, z_2, \ldots z_n$ with total thickness $Z = \Sigma z$.

(i) Flow parallel to the layers (horizontal flow)

- The hydraulic gradient i is the same in all layers; total discharge = sum of discharges.

$$k_h = (k_1 z_1 + k_2 z_2 + \ldots + k_n z_n) / (z_1 + z_2 + \ldots + z_n)$$

(ii) Flow perpendicular to the layers (vertical flow)

- The discharge velocity is the same in all layers; total head loss = sum of head losses in the layers.

$$k_v = (z_1 + z_2 + \ldots + z_n) / (z_1/k_1 + z_2/k_2 + \ldots + z_n/k_n)$$

- k_h is controlled by the most permeable layer; k_v is controlled by the least permeable layer. Always $k_h > k_v$; for natural clay deposits k_h/k_v = 2 to 10, and for varved clay up to 20 or more.
- For a single homogeneous but anisotropic soil, the equivalent isotropic permeability k_e = sqrt (k_h k_v).

Example 3

Three horizontal layers: ($k_1 = 2 \times 10^{-4}$ cm/s, z_1 = 2 m), ($k_2 = 5 \times 10^{-5}$, z_2 = 3 m), ($k_3 = 1 \times 10^{-3}$, z_3 = 1 m). Find k_h and k_v.

- $k_h = (2 \times 10^{-4} \times 2 + 5 \times 10^{-5} \times 3 + 10^{-3} \times 1) / 6 = 1.55 \times 10^{-3} / 6 = 2.58 \times 10^{-4}$ cm/s
- $k_v = 6 / (2/2 \times 10^{-4} + 3/5 \times 10^{-5} + 1/10^{-3}) = 6 / (10000 + 60000 + 1000) = 8.45 \times 10^{-5}$ cm/s
- Ratio k_h/k_v = 3.05.

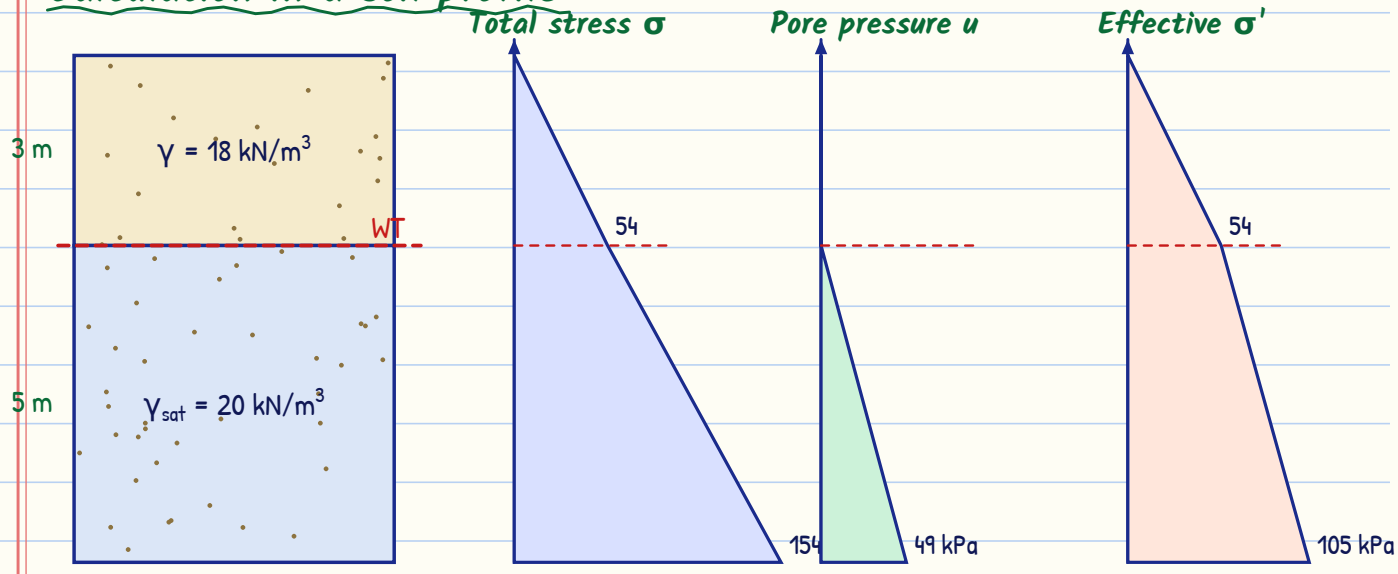
7. Total, Neutral and Effective Stresses

Terzaghi (1923) - the principle of effective stress: in a saturated soil, the total stress σ at a point is shared by the soil skeleton (effective stress σ') and the pore water (neutral stress or pore pressure u).

$$\sigma = \sigma' + u \qquad \text{or} \qquad \sigma' = \sigma - u$$

- Total stress σ = weight of everything (soil + water) above the point per unit area.
- Neutral stress u (pore water pressure) acts equally in all directions; it cannot cause deformation or shear strength. $u = \gamma_w h_w$.
- Effective stress σ' = the intergranular stress carried through the grain-to-grain contacts. All measurable effects - compression, distortion and shear strength - depend upon the effective stress, not the total stress.

Calculation in a soil profile



Sand (γ = 18 kN/m³) from 0 - 3 m; water table at 3 m; clay (γ_{sat} = 20 kN/m³) from 3 - 8 m. At 8 m depth:

- $\sigma = 18 \times 3 + 20 \times 5 = 54 + 100 = 154$ kPa
- $u = 9.81 \times 5 = 49.05$ kPa
- $\sigma' = 154 - 49.05 = 104.95$ kPa (also $\sigma' = 18 \times 3 + (20 - 9.81) \times 5 = 54 + 50.95 = 105$)

Effect of water table on effective stress

- Rising of WT reduces σ' (u increases, σ increases by smaller amount), leading to loss of strength and bearing capacity.
- Lowering of WT increases σ' and causes settlement of the ground (e.g. Mexico City, Bangkok subsidence due to pumping).

- Capillary rise above WT: u is negative, so $\sigma' = \sigma + \gamma_w h$ (increase) - apparent cohesion.
- A submerged soil: $\sigma' = z \gamma'$ (z = depth below the soil surface), independent of the water depth above the soil surface.

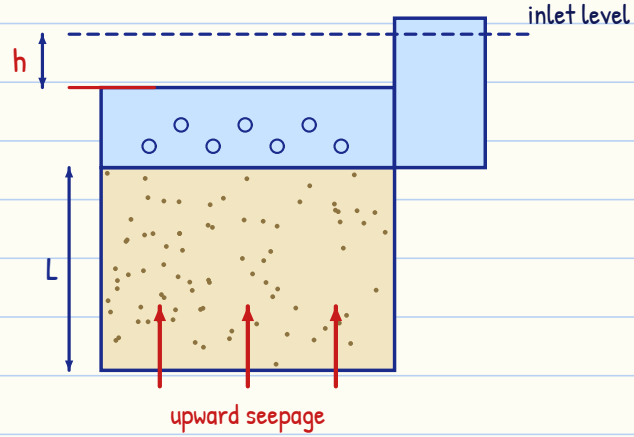
Pore pressure in a saturated soil under a static water table is hydrostatic. Water level above the ground changes both σ and u by the same amount, so σ' does not change.

8. Seepage and Seepage Pressure

When water flows through soil, it exerts a drag on the soil grains in the direction of flow, called the seepage force.

- Seepage pressure $p_s = h_L \gamma_w$ (pressure lost due to friction over the flow length) and seepage force per unit volume $j = i \gamma_w$.
- Downward seepage: σ' increases: $\sigma' = z \gamma' + i z \gamma_w$. (Helps stability.)
- Upward seepage: σ' decreases: $\sigma' = z \gamma' - i z \gamma_w = z (\gamma' - i \gamma_w)$. (Reduces stability.)

Quick Sand Condition



At the bottom of the sand layer:
total stress $\sigma = L \gamma_{sat} + z_w \gamma_w$
pore pressure $u = (L + h + z_w) \gamma_w$
$\sigma' = L \gamma' - h \gamma_w$

Quick condition: $\sigma' = 0$
$L \gamma' = h \gamma_w$
$i_c = h / L = \gamma' / \gamma_w$

As the upward hydraulic gradient increases, σ' becomes smaller. When the effective stress becomes zero, the soil grains lose contact with each other, the soil has no shear strength and behaves like a liquid - boiling of sand. This is quick sand condition (also called boiling, piping or liquefaction due to seepage).

$$\sigma' = 0 \Rightarrow i_c = \gamma' / \gamma_w = (G - 1) / (1 + e)$$

- i_c = critical hydraulic gradient; for most sands G = 2.65 and e = 0.6 - 0.8, i_c is about 1.0.
- Factor of safety against piping $F = i_c / i_{exit}$. In design F = 4 - 5 for the exit gradient (or $i_{exit} <= 0.2 - 0.25$).

- Quick sand is not a type of sand - it is a condition that can occur in any cohesionless soil (fine sand, silt) under upward flow.
- It happens: in excavations below water table, at downstream toe of dams and weirs, in river banks and cofferdams.

Remedies

- Reduce the exit gradient: longer path through cut-off walls (sheet piles), upstream impervious blanket.
- Provide filters / weighted inverted filter at the exit point, relief wells, and dewatering (wellpoints).

Example 4

A sand layer: G = 2.67, e = 0.65.

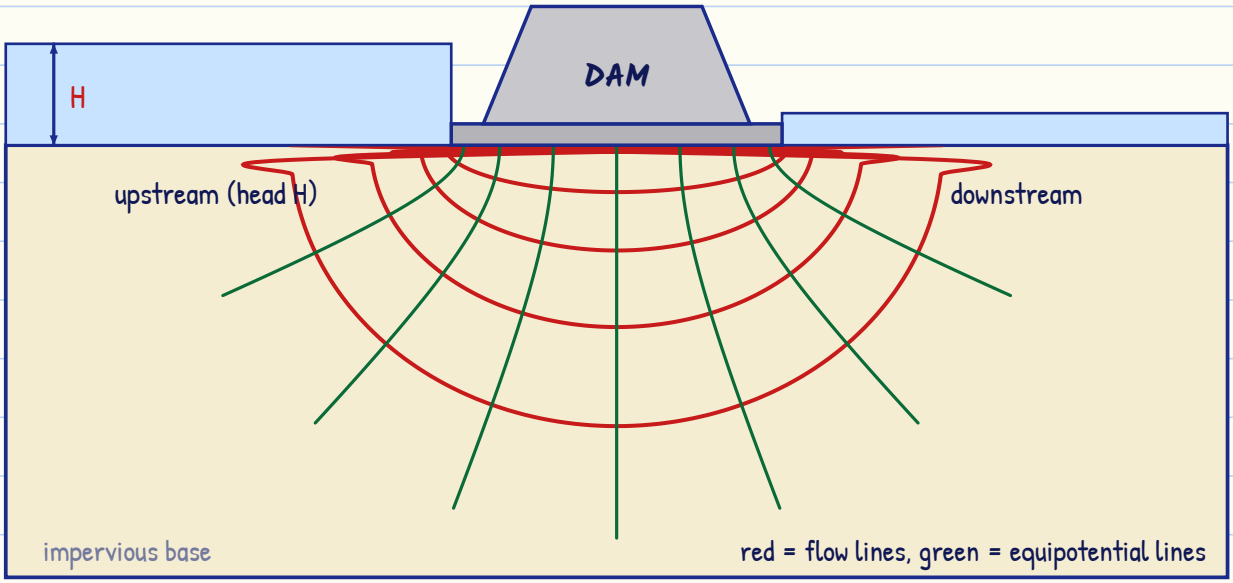
- $i_c = (2.67 - 1) / (1 + 0.65) = 1.67 / 1.65 = 1.01$.
- If a 2 m thick layer of this sand has an upward head of 1.5 m, i = 1.5/2 = 0.75, F = 1.01/0.75 = 1.35 (not safe; < 4).
- Head that would cause boiling: $h = i_c L = 1.01 \times 2 = 2.02$ m.

9. Flow Nets

Steady two-dimensional flow through isotropic soil is governed by Laplace's equation:

$$\partial^2 h/\partial x^2 + \partial^2 h/\partial z^2 = 0 \quad (\text{assuming } k_x = k_z)$$

- A flow net is the graphical solution to Laplace's equation: a net of flow lines and equipotential lines, intersecting at 90 degrees, forming curvilinear squares.
- Flow line (stream line): path followed by a water particle. Equipotential line: line joining points of equal total head.



Characteristics (properties) of a flow net

- Flow lines and equipotential lines meet at right angles (orthogonal).
- The fields (elements) formed are curvilinear squares - the average width equals the average length, so a circle can be inscribed.
- The quantity of seepage is the same in each flow channel; the head drop is the same between successive equipotentials.
- Flow lines do not cross each other; neither do equipotential lines.
- Smaller the size of a square, higher the hydraulic gradient and velocity there (e.g. near the toe).
- Impervious boundaries (base of the soil, sheet pile surface) are flow lines. Upstream and downstream soil surfaces under water are equipotential lines.

Seepage discharge

$$q = k\,H\,(N_f / N_d) \qquad \text{per unit width}$$

- H = total head loss; N_f = number of flow channels; N_d = number of equipotential drops (equipotential intervals). The ratio N_f/N_d is called the shape factor.
- Derivation: for one square field, $\Delta q = k\,(\Delta h / l) \times b = k\,\Delta h$ (since b = l); $\Delta h = H/N_d$; $q = N_f\,\Delta q$.

Pore pressure, uplift and exit gradient

- The head at the n-th equipotential: $h_n = H - n\,(H / N_d)$. Pressure head = total head - elevation head, so $u = \gamma_w\,(h_n - z)$.
- Exit gradient $i_e = (H / N_d) / l_e$, where l_e is the length of the last field at the exit. Safety against piping $F = i_c/i_e$.

- Uplift pressure on the base of a dam is obtained from the pore pressures read off the flow net.

Example 5

Flow net below a concrete dam: H = 8 m, N_f = 4, N_d = 12, k = 3 x 10^{-6} m/s.

- q = 3 x 10^{-6} x 8 x (4/12) = 8 x 10^{-6} m^3/s per metre length (= 0.69 m^3/day/m).
- Head drop per field: 8/12 = 0.667 m.
- If the last field at the toe is 1.5 m long, i_e = 0.667/1.5 = 0.44, with i_c = 1.0, F = 2.25 (not enough, add cut-off).

Construction of a flow net - steps

- Draw the structure and soil boundaries to a convenient scale. Identify boundary conditions.
- Sketch 3 - 4 flow lines, then draw equipotentials so as to get squares; adjust by trial and error until all fields are squares.
- Anisotropic soils: transform the section: $x' = x \sqrt{(k_z / k_x)}$, draw the net in the transformed section and use $k_e = \sqrt{(k_x k_z)}$ in $q = k_e H N_f/N_d$.
- Earth dams: the top flow line (phreatic line) is a basic parabola with focus at the toe of the dam (Casagrande); it is a flow line with u = 0.

Uses of flow nets

- Estimate seepage loss (quantity of seepage) under dams and through embankments.
- Determine pore water pressure at any point, hence effective stress and uplift on the base of structures.
- Find the exit gradient and check the safety against piping.
- Decide the depth of cut-off walls and the length of impervious floors.
- Check stability of slopes and design filters / drainage of earth dams.

Other methods of solution: analytical (Laplace), numerical (finite difference / finite element), electrical analogy and Casagrande's graphical method (the trial-and-error sketching method).

10. Quick Revision

Item	Formula
Capillary rise	$h_c = 4\,T\cos\alpha / (\gamma_w\, d) = C / (e\, D_{10})$
Darcy's law	$v = k\,i,\ q = k\,i\,A$
Seepage velocity	$v_s = v\,(1 + e) / e$
Constant head	$k = Q\,L / (A\,h\,t)$
Falling head	$k = 2.303\,(a\,L / A\,t)\log(h_1/h_2)$
Layered, horizontal	$k_h = \Sigma\, k\,z / \Sigma\, z$
Layered, vertical	$k_v = \Sigma\, z / \Sigma\,(z/k)$
Effective stress	$\sigma' = \sigma - u$
Critical gradient	$i_c = (G - 1)/(1 + e) = \gamma'/\gamma_w$
Flow net discharge	$q = k\,H\,N_f / N_d$

Important questions (exam)

- Explain capillary rise in soils and the effect of capillary tension on effective stress.
- State Darcy's law and its limitations. Differentiate discharge and seepage velocity.
- What are the factors affecting permeability of soils?
- Describe the constant head and falling head permeability tests, with the formulae.
- Derive the equivalent permeability for flow parallel and perpendicular to layers.
- Explain Terzaghi's effective stress principle. What is quick sand condition? Derive critical hydraulic gradient.
- What are flow nets? State the characteristics and uses; derive $q = k\,H\,N_f/N_d$.